

Joshua R. Ginsberg

West Chester, PA 19380 • 302-538-4611 • Joshginz237@gmail.com • [linkedin.com/in/joshginz](https://www.linkedin.com/in/joshginz)

Clearance: Eligible for Secret (DoD) • US Citizen (ITAR)

SUMMARY

Hands-on mechanical/manufacturing engineer with a tech-forward approach to design, test, and operations. I build practical tools, streamline operations, and deliver measurable outcomes across design, test, and digital workflow automation. Known for clear documentation, supplier alignment, and turning messy processes into simple dashboards and repeatable playbooks.

PROFESSIONAL EXPERIENCE

EDLON GMM Pfaudler, *Project Engineer*

Avondale, PA, May 2024 – Present

- Designed/released PTFE/PFA-lined vessels & piping supporting \$20M+ in orders; first-article pressure/leak tests passed on first run.
- Built parameterized FEA toolkit for vacuum-support cages/billet layouts; cut mass 45% while meeting deflection/FOS; change released via ECO.
- Drove supplier quality via source inspections (GD&T, radii/edge breaks, surface finish, weld quality); incoming nonconformances down 40% and in-plant rework down 75%.
- Digitized production tracking with SharePoint + Power Apps (standardized fields/roles, auto-revision stamps, daily exports); reduced manual entry & status-chasing, and improved on-time job closures and audit traceability.
- Set test acceptance criteria and executed pressure, thermal-vacuum, and vibration checks; produced clear results reports; hardware delivered 98% on time.

University of Delaware, *Fabrication Lab Specialist*

Newark, DE August 2022 - May 2024

- Ran the ME Design Studio (industrial 3D printers, vacuum furnace, waterjet, laser); delivered a steady flow of parts for student teams and research.
- Taught hands-on Design-for-Additive workshops; helped teams get to usable parts faster with fewer iterations.
- Repaired and returned key machines to service and documented troubleshooting steps for the next tech.
- Wrote troubleshooting SOPs and machine recovery checklists; cut downtime and made hand-offs easier.
- Standardized printer material profiles and preventive maintenance; increased print success and reduced waste.
- Operated and maintained industrial 3D printers, waterjet, laser, and vacuum former, lab uptime 98%, safety incidents 0%.

University of Delaware, *Lead Control Systems Teaching Assistant*

Newark, DE August 2023 - May 2024

- Guided students through control systems lab implementation and troubleshooting, supporting controller design/application in MATLAB/Simulink and reinforcing core theory with hands-on results.
- Developed and refined lab exercises, starter models, and course materials with faculty to improve clarity and consistency across sections, reducing recurring student errors and improving lab completion efficiency.

EDUCATION

Bachelor of Mechanical Engineering: Concentration: Aerospace

University of Delaware, Newark, DE, May 2024

Associate of Arts: General Studies

University of Delaware, Newark, DE, May 2021

TECHNICAL SKILLS

- Data Automation (Python, Excel/VBA)
- Dashboards & KPIs (SharePoint, Power Apps)
- Requirements & Systems Engineering
- Project Delivery (Agile; Jira/Azure)
- Verification & Validation (Test Plans)
- Risk & RCA (FMEA, 8D)
- PLM/BOM & Change Control
- CAD/GD&T (SolidWorks; ASME Y14.5)
- DFM/DFA & Tooling Design
- Supplier Quality (AS9100/AS9102)
- Lab & Environmental Testing
- Rapid Prototyping & 3D Printing
- Process Mapping & P&ID Documentation
- Continuous Improvement (Lean/Kaizen)
- Design of Experiments (DoE)
- Technical Writing & Stakeholder Communication